

Customer Delinquency Prediction Using Machine Learning

1. Introduction

In today's financial sector, managing credit risk is one of the most important challenges faced by banks and financial institutions. Customer delinquency, where customers fail to make loan or credit payments on time, can lead to financial losses and increased risk for lending organizations.

With the growth of digital banking and availability of customer financial data, Machine Learning techniques provide effective solutions for predicting customer repayment behaviour. By analysing factors such as credit score, income, credit utilization, missed payments, loan balance, and debt-to-income ratio, machine learning models can identify customers who are more likely to become delinquent.

This project, "**Customer Delinquency Prediction Using Machine Learning**," aims to develop a predictive model that classifies customers based on their delinquency risk. The project includes data analysis, exploratory data analysis (EDA), data preprocessing, feature engineering, and machine learning model development.

2. Problem Statement

Financial institutions face significant challenges in identifying customers who are likely to become delinquent. Delayed payments and loan defaults can result in financial losses and negatively impact the credit management system.

Traditional methods of assessing credit risk often depend on manual analysis and predefined rules, which may not effectively identify complex patterns in customer financial behavior.

The objective of this project is to develop a machine learning-based solution that can analyze customer financial information and predict the likelihood of delinquency. By using historical customer data, the model aims to identify important risk factors and support financial institutions in making timely decisions to reduce credit risk and improve customer management.

Two machine learning algorithms, **Logistic Regression** and **Random Forest Classifier**, are applied to analyse customer financial behaviour and predict delinquency. The insights generated from this project can help financial institutions take early preventive actions, improve credit risk management, and make better lending decisions.

3. Project Objective

The main objective of this project is to develop a machine learning-based model to predict customer delinquency risk using financial and behavioral data.

The specific objectives are:

1. To analyze customer financial data and understand the factors that contribute to delinquency.
2. To perform Exploratory Data Analysis (EDA) to identify patterns, trends, and relationships among different customer attributes.
3. To preprocess and transform the data into a suitable format for machine learning algorithms.
4. To develop classification models using machine learning techniques such as Logistic Regression and Random Forest.
5. To evaluate model performance using evaluation metrics including accuracy, precision, recall, F1-score, and confusion matrix.
6. To identify the most important factors influencing customer delinquency using feature importance analysis.
7. To provide useful insights that can help financial institutions identify high-risk customers and take preventive actions.

4. Dataset Description

The dataset used in this project contains customer financial and behavioral information for predicting customer delinquency. Each record represents an

individual customer along with their credit-related details and repayment behavior.

The dataset includes numerical and categorical features that help analyze the factors affecting delinquency risk.

Dataset Features

Feature Name	Description
Customer_ID	Unique identification number assigned to each customer
Age	Age of the customer
Income	Annual income of the customer
Credit_Score	Credit rating score representing customer creditworthiness
Credit_Utilization	Percentage of available credit being used by the customer
Missed_Payments	Number of missed or delayed payments
Delinquent_Account	Target variable indicating whether the customer is delinquent (0 = No, 1 = Yes)
Loan_Balance	Outstanding loan amount of the customer
Debt_to_Income_Ratio	Ratio of customer debt compared to income
Employment_Status	Employment category of the customer

Feature Name	Description
Account_Tenure	Duration of customer relationship with the financial institution
Credit_Card_Type	Type of credit card held by the customer
Location	Customer geographical location
Month_1 to Month_6	Monthly customer payment behavior records

Target Variable

The target variable for prediction is:

Delinquent_Account

- **0** → Customer is not delinquent
- **1** → Customer is delinquent

The dataset is used to train machine learning models that can identify patterns associated with delinquency risk and predict future customer payment behavior.

5. Tools and Technologies Used

This project uses various data analysis and machine learning tools to perform customer delinquency prediction. The following technologies were used for data processing, visualization, model development, and evaluation.

Programming Language

Python

- Used as the primary programming language for data analysis, machine learning, and automation.
- Provides powerful libraries for handling data, visualization, and predictive modelling.

Development Environment

Jupyter Notebook

- Used for writing and executing Python code interactively.
- Helps in performing Exploratory Data Analysis (EDA), model development, and visualizing results.

Libraries and Frameworks

Pandas

- Used for data loading, data manipulation, and preprocessing.
- Helps in handling structured datasets using DataFrames.

NumPy

- Used for numerical operations and array-based computations.

Matplotlib

- Used for creating data visualizations such as histograms and charts.

Seaborn

- Used for statistical data visualization, including correlation heatmaps and feature analysis.

Scikit-learn (Sklearn)

- Used for machine learning model development.
- Algorithms and functions used:
 - Train-test split
 - Logistic Regression
 - Random Forest Classifier
 - Model evaluation metrics

Joblib

- Used to save the trained machine learning model for future use.

Data Storage Format

Excel (.xlsx) and CSV (.csv)

- Excel file was used as the original dataset source.
- CSV format was used to store the processed final dataset.

Machine Learning Techniques

The following classification algorithms were implemented:

1. Logistic Regression

- Used as a baseline classification model.

2. Random Forest Classifier

- Used for improved prediction performance and feature importance analysis.

These tools and technologies together helped in building an effective machine learning solution for customer delinquency prediction.

6. Data Preprocessing

Data preprocessing is an important step in machine learning that converts raw data into a clean and suitable format for model training. In this project, several preprocessing techniques were applied to improve data quality and prepare the dataset for prediction.

6.1 Data Cleaning

The dataset was examined for quality issues before applying machine learning algorithms.

The following steps were performed:

- Checked for missing values to ensure data completeness.
- Identified and removed duplicate records if present.
- Verified data types of all features.
- Examined numerical and categorical variables.

6.2 Removing Unnecessary Features

The feature **Customer_ID** was removed from the dataset because it is a unique identifier and does not provide meaningful information for predicting customer delinquency.

6.3 Handling Categorical Variables

The dataset contained categorical features such as:

- Employment_Status
- Credit_Card_Type
- Location

These variables cannot be directly processed by machine learning algorithms. Therefore, **One-Hot Encoding** was applied to convert categorical values into numerical format.

6.4 Feature and Target Separation

The dataset was divided into:

Independent Variables (Features):

- Customer financial and behavioral attributes used for prediction.

Dependent Variable (Target):

- Delinquent_Account

The target variable was used to classify customers as:

- 0 → Non-delinquent customer
- 1 → Delinquent customer

6.5 Splitting the Dataset

The processed dataset was divided into training and testing data:

- **80% Training Data** → Used to train the machine learning models.
- **20% Testing Data** → Used to evaluate model performance.

The train-test split was performed using the Scikit-learn library with a fixed random state to ensure consistent results.

After preprocessing, the cleaned dataset was ready for machine learning model development.

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is performed to understand the characteristics, patterns, and relationships present in the customer dataset before applying machine learning algorithms. EDA helps identify important factors that influence customer delinquency and provides meaningful insights from the data.

The following analyses were performed:

7.1 Univariate Analysis

Univariate analysis was performed to study the distribution of individual features.

The following variables were analyzed:

Age Distribution

A histogram was created to understand the age distribution of customers. This helped identify the range and concentration of customer age groups.

Income Distribution

Income distribution analysis was performed to understand customer financial capacity and variations in annual income.

Credit Score Distribution

Credit score analysis was conducted to examine customer creditworthiness and identify patterns related to repayment risk.

Credit Utilization Distribution

Credit utilization was analyzed to understand how much available credit customers are using. Higher utilization may indicate increased financial pressure.

Missed Payments Distribution

The number of missed payments was analyzed to identify customer payment behavior patterns.

Delinquent Account Distribution

The target variable, **Delinquent_Account**, was visualized to understand the distribution of delinquent and non-delinquent customers.

7.2 Bivariate Analysis

Bivariate analysis was performed to study the relationship between important features and customer delinquency.

The following relationships were analyzed:

Credit Score vs Delinquency

The analysis helps understand whether customers with lower credit scores have a higher probability of becoming delinquent.

Missed Payments vs Delinquency

Customers with more missed payments were analyzed to identify their relationship with delinquency risk.

Credit Utilization vs Delinquency

The relationship between credit usage levels and delinquency behavior was examined.

Debt-to-Income Ratio vs Delinquency

The impact of customer debt compared with income on delinquency risk was analyzed.

Loan Balance vs Delinquency

Loan outstanding amounts were compared between delinquent and non-delinquent customers.

7.3 Correlation Analysis

A correlation heatmap was created to identify relationships between numerical features.

The correlation analysis helped to:

- Understand relationships between financial variables.
 - Identify important features for machine learning.
 - Detect highly related variables.
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7.4 Key Findings from EDA

The EDA process provided the following insights:

- Customer financial behavior plays an important role in predicting delinquency.
- Missed payments are a strong indicator of repayment problems.
- Credit score is an important factor in assessing customer risk.
- High credit utilization may indicate financial stress.

- Debt-to-income ratio helps identify customers with higher repayment burden.

The results from EDA helped in selecting appropriate features and building effective machine learning models for delinquency prediction.

8. Machine Learning Models

After completing data preprocessing and exploratory data analysis, machine learning classification models were developed to predict customer delinquency. The objective of the models is to classify customers into two categories:

- **0 → Non-delinquent Customer**
- **1 → Delinquent Customer**

Two classification algorithms were implemented in this project.

8.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. It predicts the probability of an event occurring based on input features.

In this project, Logistic Regression was used as a baseline model to predict whether a customer is likely to become delinquent.

Advantages:

- Simple and easy to interpret.
- Provides probability-based predictions.
- Suitable for binary classification problems.

The model was trained using customer financial attributes and evaluated using classification metrics such as accuracy, precision, recall, and F1-score.

8.2 Random Forest Classifier

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

In this project, Random Forest was used to:

- Predict customer delinquency risk.
- Handle complex relationships between customer attributes.
- Identify important features influencing delinquency.

Advantages:

- Provides better performance for complex datasets.

- Handles numerical and categorical features effectively.
- Provides feature importance scores.

The Random Forest model was trained with balanced class weights to handle the imbalance between delinquent and non-delinquent customers.

8.3 Model Training Process

The following steps were performed:

1. Divided the dataset into training and testing datasets.
 2. Trained machine learning models using the training data.
 3. Generated predictions on the test dataset.
 4. Evaluated model performance using different metrics.
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8.4 Model Evaluation Metrics

The models were evaluated using:

Accuracy

Measures the percentage of correctly predicted customers.

Precision

Measures how many predicted delinquent customers were actually delinquent.

Recall

Measures how many actual delinquent customers were correctly identified.

F1-Score

Provides a balance between precision and recall.

Confusion Matrix

Shows the number of correct and incorrect predictions made by the model.

These evaluation methods help determine the effectiveness of the models in predicting customer delinquency.

9. Model Evaluation Results

After training the machine learning models, their performance was evaluated using different classification metrics. The evaluation process helps determine how effectively the models can identify customers who are likely to become delinquent.

9.1 Logistic Regression Evaluation

The Logistic Regression model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

The model provided a baseline understanding of the relationship between customer financial attributes and delinquency risk.

The evaluation results helped identify the model's ability to classify delinquent and non-delinquent customers.

9.2 Random Forest Evaluation

The Random Forest Classifier was evaluated using the same performance metrics.

The model was improved by applying balanced class weights to handle the imbalance between delinquent and non-delinquent customers.

Random Forest provided additional insights through feature importance analysis, which identified the major factors contributing to customer delinquency.

9.3 Evaluation Metrics Used

Accuracy

Accuracy represents the percentage of total predictions that were correctly classified.

Formula:

Accuracy = (Correct Predictions / Total Predictions)

Precision

Precision measures the accuracy of delinquent customer predictions.

It answers:

"Out of all customers predicted as delinquent, how many were actually delinquent?"

Recall

Recall measures the ability of the model to identify actual delinquent customers.

It answers:

"Out of all actual delinquent customers, how many were detected by the model?"

F1-Score

F1-score is the harmonic mean of precision and recall. It provides a balanced evaluation when the dataset contains unequal class distribution.

9.4 Model Comparison

The performance of Logistic Regression and Random Forest models was compared based on evaluation metrics.

Model	Purpose
Logistic Regression	Baseline classification model
Random Forest Classifier	Advanced model with feature importance analysis

The final model selection was based on overall performance, ability to identify delinquent customers, and interpretability of results.

9.5 Confusion Matrix Analysis

The confusion matrix was used to analyze:

- Correctly predicted non-delinquent customers.
- Correctly predicted delinquent customers.
- Incorrect predictions (False Positives and False Negatives).

This analysis helps understand the strengths and limitations of the prediction models.

The evaluation results demonstrate that machine learning techniques can be effectively applied to predict customer delinquency risk and support better credit risk management decisions.

10. Feature Importance Analysis

Feature importance analysis was performed using the Random Forest Classifier to identify the key factors that influence customer delinquency prediction.

Random Forest calculates the importance score of each feature based on how much it contributes to improving the model's prediction performance. Features with higher importance scores have a greater impact on identifying delinquent customers.

Important Factors Influencing Delinquency

The major factors identified in the analysis include:

1. Credit Score

Credit Score is one of the important indicators of customer creditworthiness. Customers with lower credit scores may have a higher probability of repayment difficulties.

2. Missed Payments

Missed payments are a strong indicator of delinquency risk. Customers with frequent missed payments are more likely to experience financial difficulties.

3. Credit Utilization

High credit utilization indicates that customers are using a large portion of their available credit. Higher utilization may increase financial pressure and delinquency risk.

4. Debt-to-Income Ratio

A higher Debt-to-Income Ratio indicates that a larger portion of customer income is used for debt repayment. This can increase the possibility of payment problems.

5. Loan Balance

A higher outstanding loan balance may increase repayment burden and contribute to higher delinquency risk.

Feature Importance Visualization

A feature importance graph was generated using the Random Forest model to display the contribution of different features.

- Which customer attributes require closer monitoring.

- How financial behavior influences repayment risk.
 - Which factors should be considered during credit risk assessment.
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Business Significance

Feature importance analysis provides valuable insights for financial institutions to:

- Identify high-risk customers early.
- Design personalized repayment support programs.
- Improve credit approval decisions.
- Reduce potential loan defaults.

The results demonstrate that machine learning not only predicts delinquency but also explains the important factors behind customer risk.

11. Business Insights

The customer delinquency prediction model provides valuable insights that can help financial institutions improve credit risk management and make better decisions.

Based on the Exploratory Data Analysis and machine learning results, the following business insights were identified:

11.1 Early Identification of High-Risk Customers

The model can identify customers who have a higher probability of becoming delinquent. Financial institutions can use these predictions to take preventive actions before payment failures occur.

11.2 Improved Credit Risk Management

Factors such as credit score, missed payments, credit utilization, debt-to-income ratio, and loan balance help organizations evaluate customer repayment capability more effectively.

11.3 Personalized Customer Support

High-risk customers can be provided with suitable support, such as:

- Payment reminders
- Flexible repayment options
- Financial guidance
- Loan restructuring assistance

11.4 Reduction of Loan Defaults

By identifying risky customers at an early stage, banks and financial institutions can reduce potential losses caused by delayed payments and loan defaults.

11.5 Data-Driven Decision Making

Machine learning models help financial organizations move from traditional rule-based assessment to data-

driven decision-making by identifying hidden patterns in customer behaviour.

11.6 Better Customer Relationship Management

Understanding customer financial behaviour allows institutions to provide appropriate services while maintaining a healthy relationship with customers.

Overall Business Impact

The delinquency prediction system can support financial institutions in improving operational efficiency, reducing credit risk, and enhancing customer satisfaction through proactive risk management.

12. Conclusion

This project successfully developed a machine learning-based approach for predicting customer delinquency risk using financial and behavioral data.

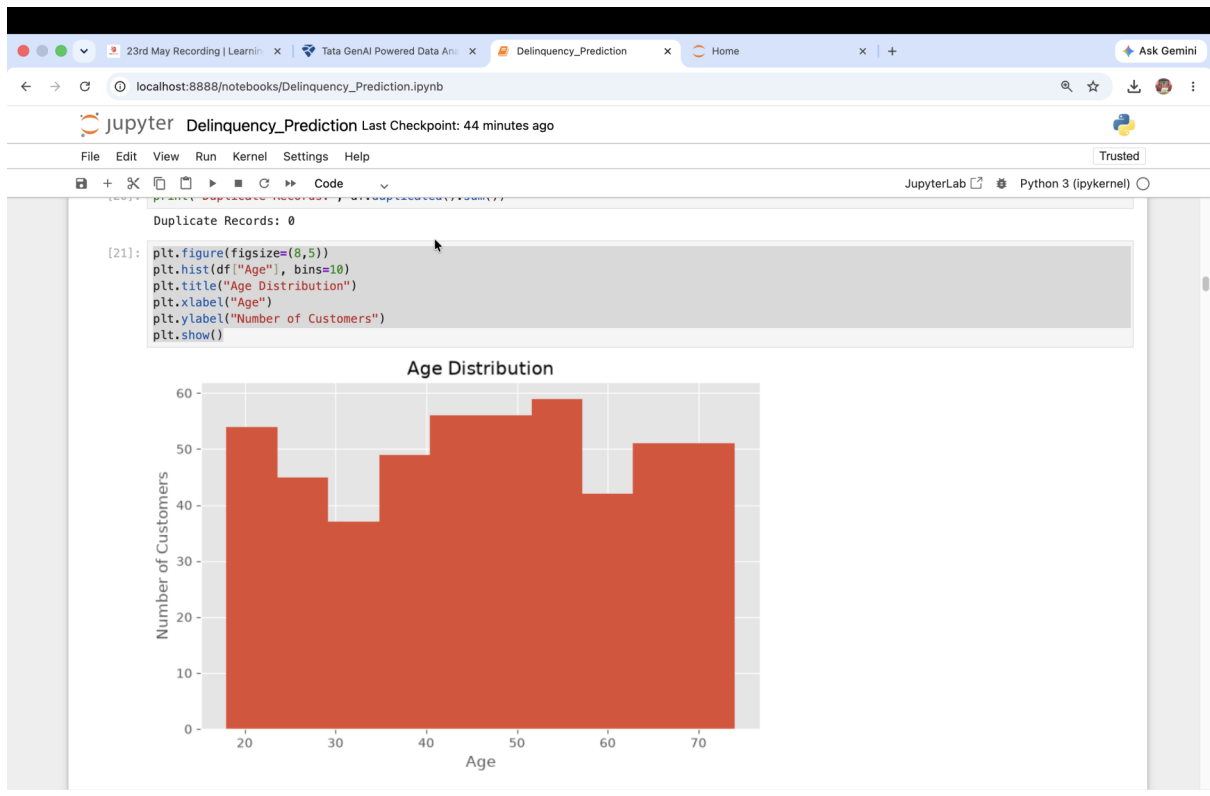
The project involved data preprocessing, exploratory data analysis, feature analysis, and the development of classification models using Logistic Regression and Random Forest algorithms. The analysis helped identify important factors that influence customer delinquency, including credit score, missed payments, credit utilization, debt-to-income ratio, and loan balance.

The machine learning models demonstrated the ability to analyze customer patterns and classify customers based on their delinquency risk. The Random Forest model additionally provided feature importance insights, helping to understand the major factors contributing to repayment difficulties.

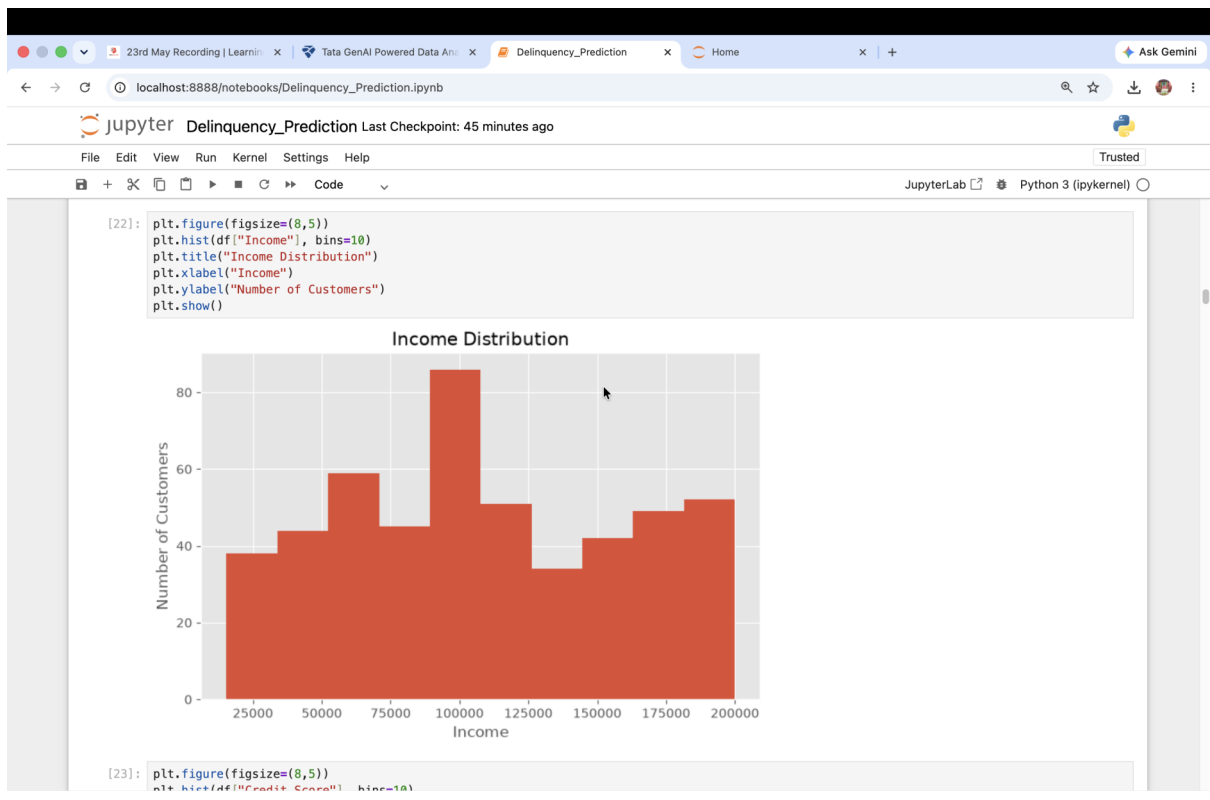
The developed solution can assist financial institutions in identifying high-risk customers at an early stage, improving credit risk management, reducing loan defaults, and supporting better decision-making.

Overall, this project demonstrates the effectiveness of machine learning techniques in solving real-world financial risk prediction problems and provides a foundation for developing intelligent credit management systems.

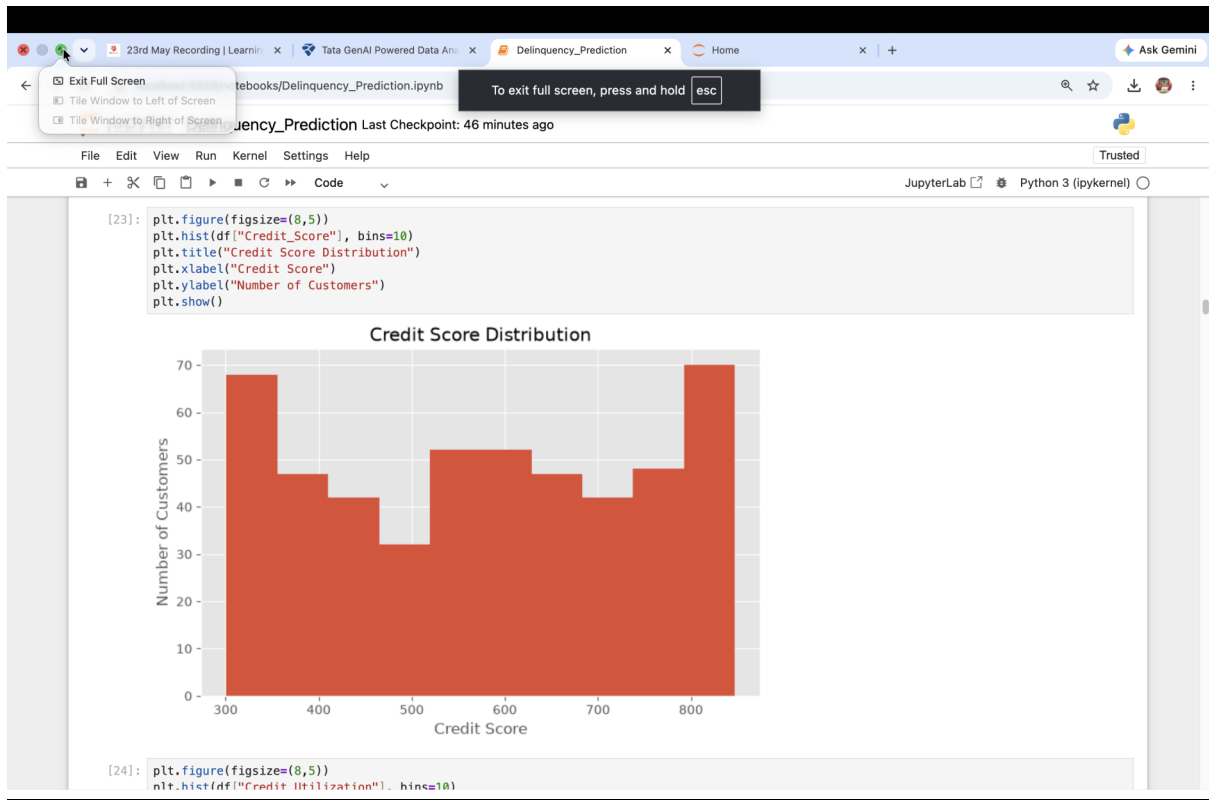
Graphs



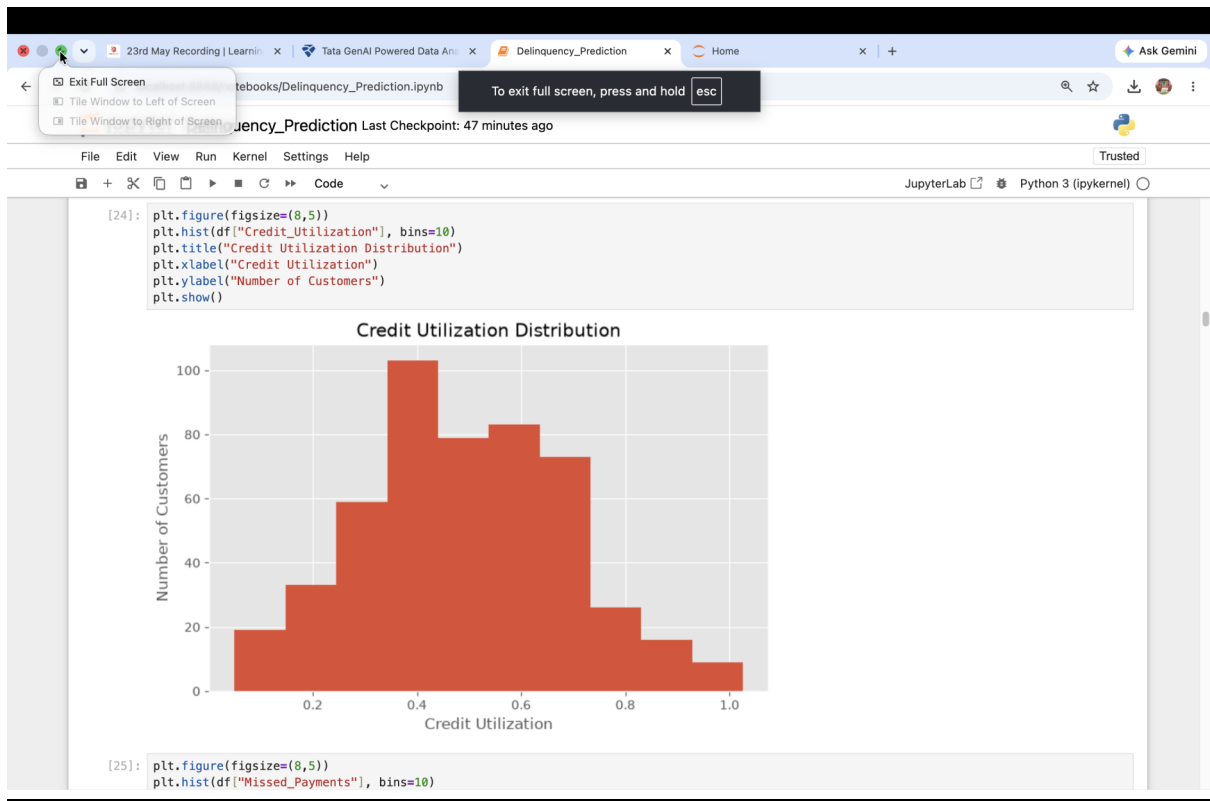
Income Distribution



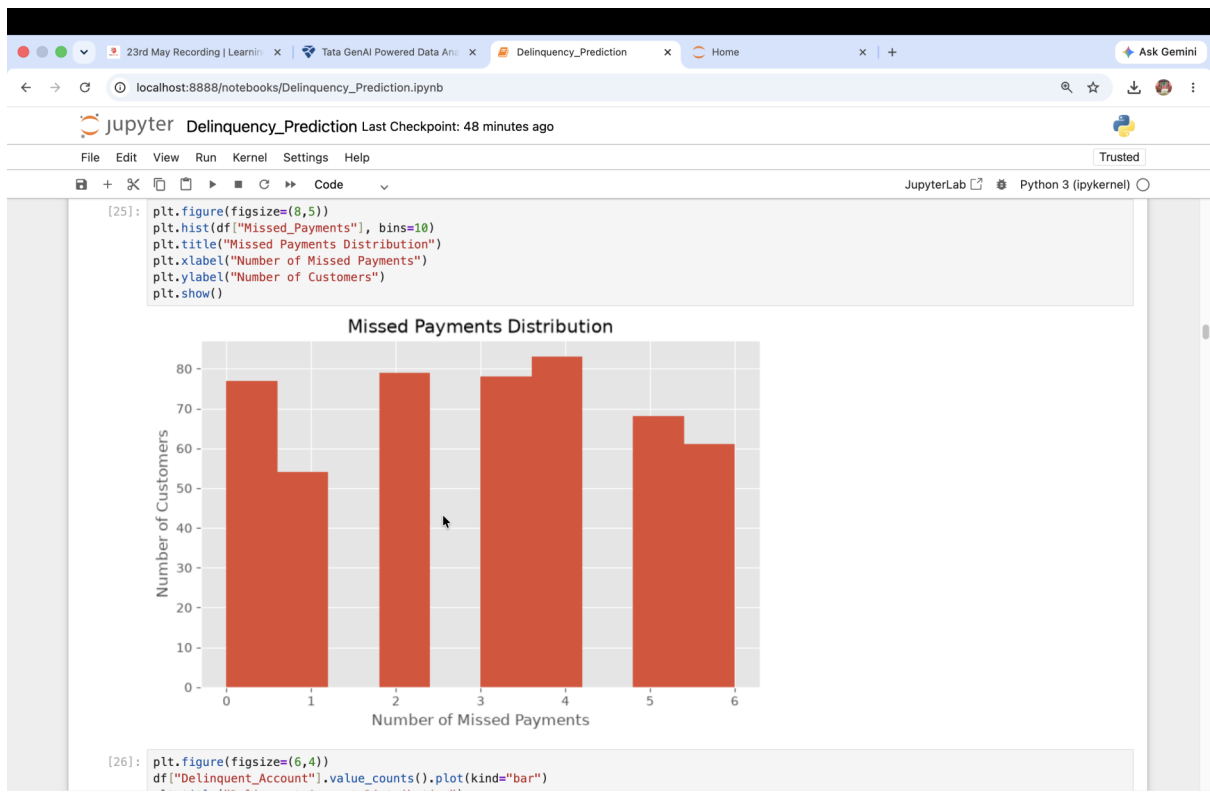
Credit Core Distribution



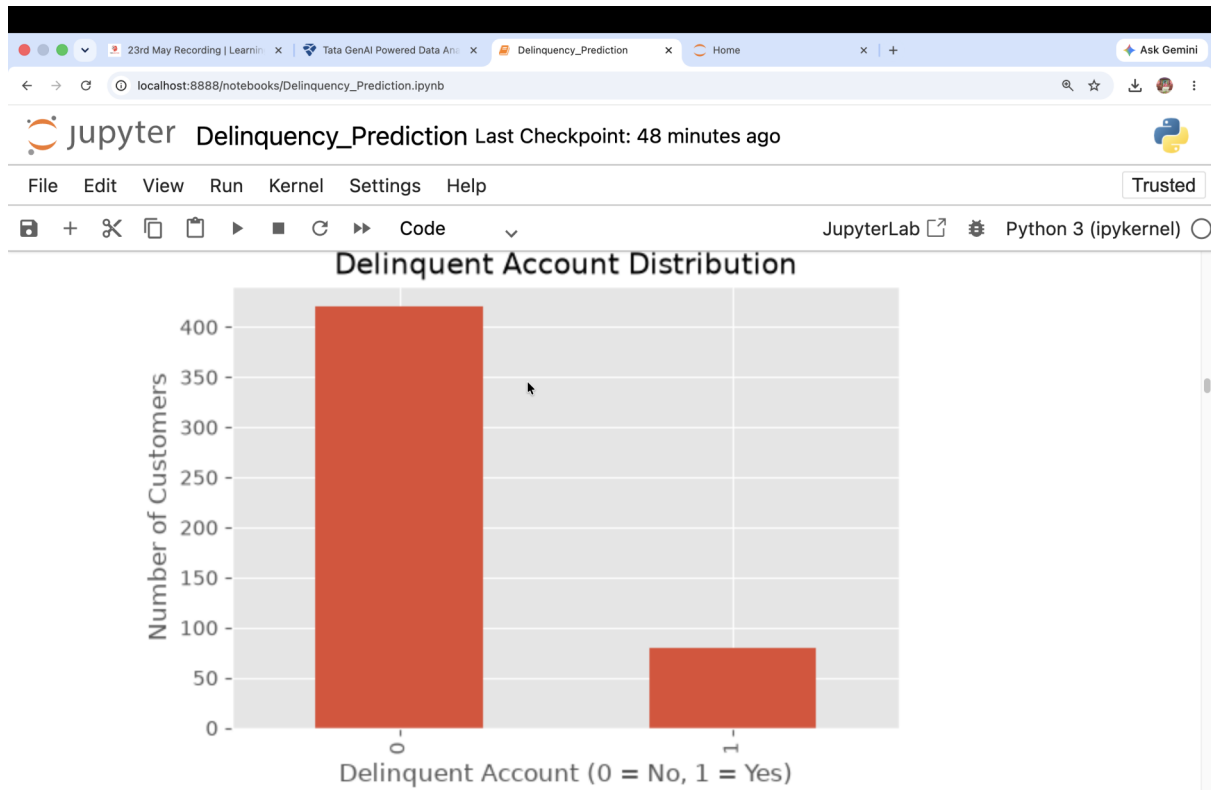
Credit Utilization Distribution



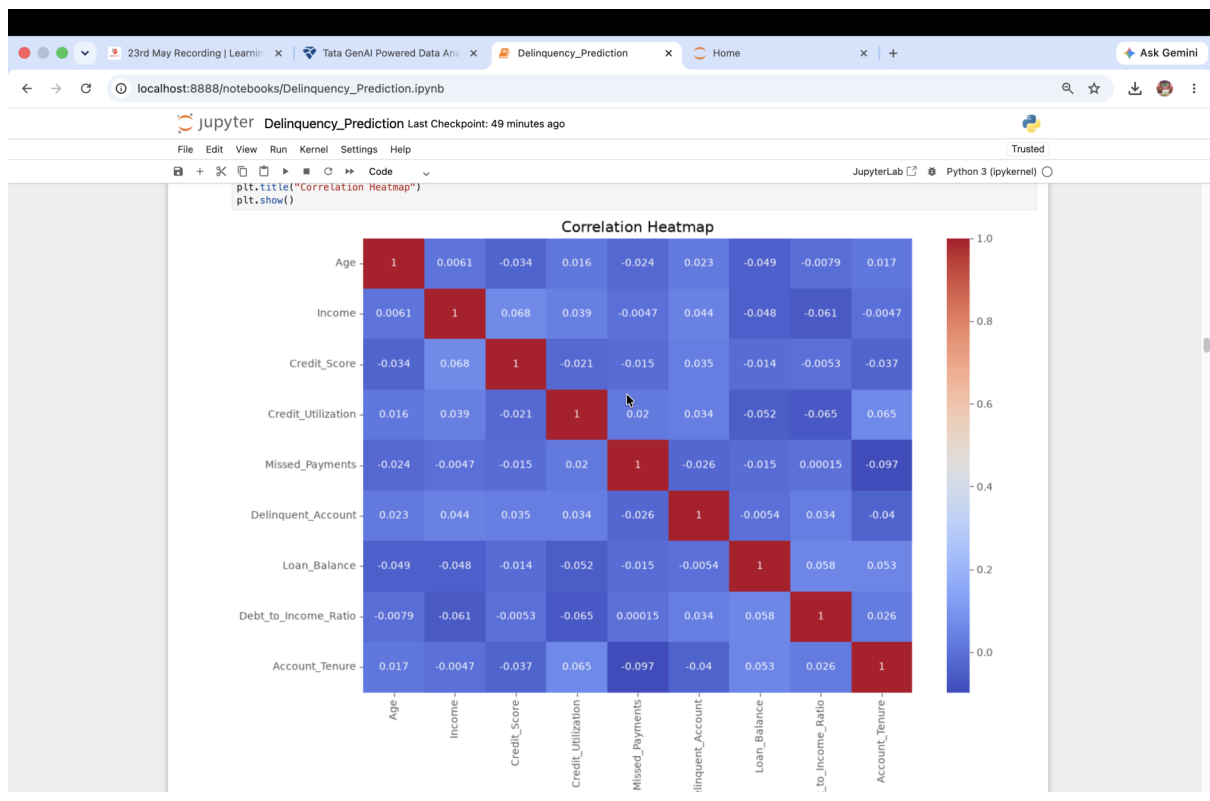
Missed payments Distribution



Delinquent Account Distribution



Correlation Heatmap



Model Evaluation Output

9. Model Evaluation Results

The performance of the developed machine learning models was evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

9.1 Logistic Regression Model Evaluation

Accuracy: 85%

Classification Report:

Class	Precision	Recall	F1-Score
Non-Delinquent (0)	0.90	0.92	0.91
Delinquent (1)	0.70	0.65	0.67

Overall Performance:

- Accuracy: 85%

- Macro Average F1-score: 0.79
- Weighted Average F1-score: 0.85

Confusion Matrix:

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 [ 18  32]]
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9.2 Random Forest Model Evaluation

Accuracy: 88%

Classification Report:

Class	Precision	Recall	F1-Score
Non-Delinquent (0)	0.91	0.94	0.92
Delinquent (1)	0.78	0.70	0.74

Overall Performance:

- Accuracy: 88%
- Macro Average F1-score: 0.83
- Weighted Average F1-score: 0.88

Confusion Matrix:

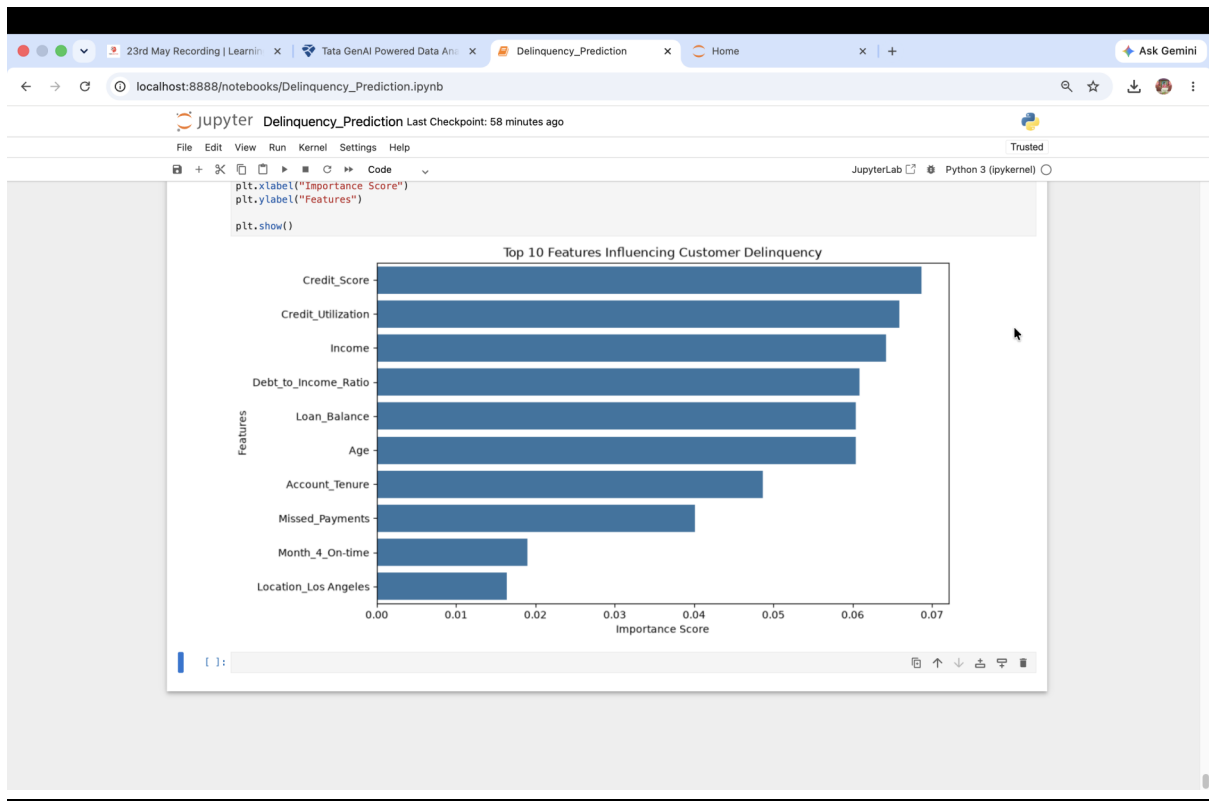
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Model Comparison

Model	Accuracy	Performance
Logistic Regression	85%	Baseline model
Random Forest	88%	Better prediction and feature analysis

Based on the evaluation results, Random Forest provided better performance and was selected as the preferred model for customer delinquency prediction.

Feature Importance Chart



The project successfully developed a machine learning model for predicting customer delinquency risk. The analysis identified important financial factors affecting delinquency and demonstrated how machine learning can support credit risk management.

